



Name: Kakumanu Rajesh Kumar

Role: Senior Backend Engineer (Python / Polyglot Microservices / AWS)

Total Experience: 8 years

Phone No: +91 96666 99540

Email: kakumanurajeshkumar@gmail.com

Working At: TAO Digital Solutions

SUMMARY

Senior Software Engineer with **8+ years of experience** specializing in production **Python** and polyglot microservices (**FastAPI, Node.js/NestJS**) on scalable **AWS serverless (Lambda)** and cloud architectures. Expert in **event-driven systems**, robust API security (**Auth0, OAuth 2.0/OIDC, JWT, RBAC**), and high-throughput data layers using **MongoDB (MongoEngine)** and SQL. Proven track record leading technical designs, containerizing local and CI/CD environments (**Docker, GitHub Actions**), and partnering with **React and Vue.js** frontend teams on strict API contracts. Pioneer in **AI-native engineering (LangGraph, LangChain, MCP)** to accelerate delivery velocity while maintaining enterprise-grade test and code quality.

TECHNICAL PROFICIENCIES

- **Backend & Architecture:** Python (FastAPI, Flask, Django), Node.js, NestJS, Express.js, Microservices, Event-Driven Architecture, RESTful API Design & Versioning, GraphQL, Linux/Unix CLI.
- **Databases & ORM/ODM:** MongoDB, MongoEngine, PostgreSQL, MySQL, SQLAlchemy, Redis (Distributed Caching), DynamoDB, Schema Design & Query Optimization.
- **Authentication & Security:** Auth0, OAuth 2.0 / OIDC, SAML, JWT Validation & Token Lifecycle Management, Role-Based Access Control (RBAC), API Gateway Authorizers.
- **Cloud & Serverless (AWS):** AWS Lambda, API Gateway, EventBridge, SQS, SNS, CloudWatch, S3, EC2, Secrets Manager, Cognito.
- **DevOps, CI/CD & Containers:** Docker (Local Development & Multi-Stage Builds), GitHub Actions, CI/CD Automated Pipelines, Linux/Unix Shell Scripting, Docker Compose.
- **AI & Agentic Engineering:** Model Context Protocol (MCP), LangGraph, LangChain, Claude Code, GitHub Copilot (Agent Mode), Agentic Workflows, Prompt & Context Engineering.
- **Data Science & ML Stack:** PyTorch, TensorFlow (RNN, LSTM, GRU), NumPy, Pandas, Scikit-Learn, Computation Graphs & Vectorized Operations.
- **Testing & Quality Assurance:** Pytest, Jest, Playwright MCP (AI-Driven Testing), API Contract Testing, Self-Healing Tests, Integration & Unit Test Automation.
- **Frontend & Collaboration:** React.js, Vue.js, Next.js, TypeScript, Tailwind CSS, State Management (Redux/Pinia/Context API), OpenAPI/Swagger Contract Testing.

PROFESSIONAL EXPERIENCE

TAO Digital Solutions | Senior Software Engineer | Oct 2023 – Present

Continuous Upskilling & Applied AI R&D | June 2026 – Present

Stack: Python (FastAPI), LangGraph, LangChain, Node.js/NestJS, AWS (Lambda, EventBridge, SQS/SNS), MongoDB, GitHub Copilot, MCP

- **Stateful Multi-Agent Systems:** Deepened hands-on expertise in **LangGraph** and **LangChain** by architecting cyclical agent runtimes, implementing state persistence, dynamic context-stitching, and self-correcting validation loops for deterministic decision-making.
- **Serverless & Event-Driven Tooling:** Prototyped event-driven microservices integrating **FastAPI** with **AWS Lambda**, **SQS/SNS**, and **EventBridge** to simulate asynchronous agent task dispatching and high-concurrency event handling.
- **Agentic Developer Workflows:** Researched and integrated **Model Context Protocol (MCP)** with GitHub Copilot (Agent Mode) to experiment with autonomous code refactoring and self-healing test generation.

Project: Agentic Report Validation (ARV) — MVP Platform | Client: Oaktree Capital Management | June 2026 – August 2026

Role: Lead Python Backend / Agentic AI Engineer

Stack: Python (FastAPI), AWS (Lambda, API Gateway, EventBridge, SQS/SNS, CloudWatch), MongoDB (MongoEngine), Docker & Docker Compose, GitHub Actions (CI/CD), Auth0 (OAuth 2.0, JWT, RBAC), Pytest, Vue.js & React, Linux/Unix, Anthropic Claude

- **Production Python MVP & Document Store:** Architected and launched the production MVP for a multi-agent financial validation engine in **Python (FastAPI)**. Designed a flexible document-store layer using **MongoDB** with **MongoEngine** to persist heterogeneous extraction schemas, dynamic parameter definitions, and agent memory state, implementing compound indexing and query profiling to sustain sub-second lookups across quarterly datasets.
- **Serverless Compute & Event-Driven Backbone:** Engineered an asynchronous, 7-agent pipeline (Ingestion, Layout, Extraction, Variance, Business Rules, QA, Review) across ~21k LOC of production Python. Decoupled high-concurrency ingestion and heavy parsing jobs using **AWS EventBridge** and **SQS/SNS**, deploying **AWS Lambda** functions behind **API Gateway** for lightweight serverless preprocessing and file normalization.
- **Containerization & CI/CD Pipeline Engineering:** Containerized the entire backend using multi-stage **Docker** builds and configured **Docker Compose** for parity between local

development and production. Automated linting, build verification, and contract suites via **GitHub Actions**, enforcing automated quality and security checks on every pull request.

- **REST API Design, Versioning & UI Contract Collaboration:** Designed, built, and versioned multi-tenant REST APIs, partnering closely with frontend engineers to define OpenAPI schemas consumed across **Vue.js** dashboard views and **React** report review workflows. Secured endpoints via **Auth0**, implementing **OAuth 2.0/OIDC** flows, strict **JWT** lifecycle and signature validation, and granular **RBAC** for document tier access.
- **Extraction Intelligence & Cost-Gated Execution:** Built an autonomous 5-stage document extraction ladder (pdfplumber ruled-line → stream → word-bbox → character-cell clustering → Claude Vision), achieving 278/278 deal recoveries from borderless statements. Implemented a 3-mode execution engine (rule/dynamic/agentic) with durable episodic memory, keeping inference cost opt-in and zero-token by default.
- **Automated Contract Testing & Guardrails:** Enforced API contracts and extraction integrity with **638+ Pytest unit and integration tests** and an 820-metric regression diff gating releases. Automated parameter validation within the **GitHub Actions** CI pipeline to reject builds on unmapped financial parameters, expanding validated coverage from 18 to 37.

Project: Neuron Performance Suite (NPS) | Client: CDK Global | February 2025 – June 2026

Role: Senior Backend Engineer (Python & Polyglot Microservices)

Stack: Python, NestJS, Next.js, Vue.js, MongoDB (MongoEngine), PostgreSQL, Docker & Docker Compose, GitHub Actions (CI/CD), AWS (Lambda, API Gateway, EventBridge, SQS/SNS, CloudWatch), Auth0, Pytest, Jest, Linux/Unix, MCP

- **Production Python & Microservices Architecture:** Co-architected a high-availability microservices ecosystem leveraging **Python** and NestJS, designing, building, and versioning resilient REST APIs consumed by multi-tenant automotive and IoT telemetry clients. Decoupled high-throughput workloads using an event-driven backbone powered by **AWS EventBridge**, **SQS**, and **SNS**.
- **High-Performance Document Modeling (MongoDB):** Architected and optimized scalable document schemas in **MongoDB** using **MongoEngine**, implementing compound indexing strategies, aggregation pipelines, and query profiling that slashed read latencies by 40% across high-velocity diagnostic and IoT data streams.
- **Serverless Compute & Enterprise IAM:** Engineered production **AWS Lambda** serverless functions behind **API Gateway** to process asynchronous data transformations and webhook triggers. Integrated enterprise authentication via **Auth0**, enforcing **OAuth 2.0/OIDC** flows, **JWT** signature validation, dynamic token lifecycle handling, and fine-grained **RBAC**.
- **Containerization & CI/CD Pipeline Engineering:** Standardized local development and staging environments using multi-stage **Docker** configurations and **Docker Compose**, eliminating environment drift across polyglot services. Built automated **GitHub Actions** CI/CD pipelines to enforce linting, image vulnerability scans, and deployment gates.

- **API Contract Testing & Frontend Collaboration:** Formulated strict OpenAPI/Swagger contracts and automated consumer-driven contract tests, collaborating cross-functionally with **Vue.js** and **React/Next.js** frontend teams to eliminate integration blockers. Maintained >85% code coverage through automated unit and integration suites using **Pytest** and Jest.
- **Agentic Workflows & System Observability:** Pioneered the use of **Model Context Protocol (MCP)** with GitHub Copilot (Agent Mode) and Playwright MCP to automate E2E test generation and refactoring, improving test stability by 35%. Monitored end-to-end service health, traces, and metrics using **AWS CloudWatch** and automated **Linux/Unix** diagnostic scripts.

Project: **twiist™ AID System** | **Client:** **Sequel**

Role: Senior Backend Engineer (Python, Cloud & Serverless)

Stack: Python, NestJS, AWS (Lambda, API Gateway, EventBridge, SQS, SNS, SES, CloudWatch), MongoDB (MongoEngine), PostgreSQL/Sequelize, Docker & Docker Compose, GitHub Actions, Auth0 & AWS Cognito (OAuth 2.0, JWT, RBAC), Pytest, Vue.js, Linux/Unix

- **Healthcare IoT & Event-Driven Orchestration:** Co-architected an event-driven data ingestion platform connecting automated insulin delivery (AID) devices with external platforms (Tidepool, Salesforce, MuleSoft). Built asynchronous ingestion pipelines using **AWS EventBridge**, **SQS**, and **SNS** to reliably dispatch high-frequency glycemic telemetry and device alerts.
- **Serverless Compute & Document Ingestion:** Engineered lightweight, high-throughput **AWS Lambda** microservices in **Python** behind **API Gateway** to parse and normalize continuous glucose monitor (CGM) streams. Modeled semi-structured patient telemetry and device logs in **MongoDB** using **MongoEngine**, designing time-series indexing and aggregation queries for low-latency clinical dashboards.
- **HIPAA Compliance, Identity & API Security:** Architected HIPAA-compliant REST APIs enforcing end-to-end data encryption, audit trails, and strict enterprise IAM. Implemented secure access boundaries utilizing **Auth0** and **Cognito**, managing **OAuth 2.0/OIDC** grant types, **JWT** signature validation, token refresh lifecycles, and fine-grained **RBAC** for patients, caregivers, and clinicians.
- **Dockerized Workflows & CI/CD Pipelines:** Containerized microservices using **Docker** and multi-container **Docker Compose** setups for reproducible local development across **Linux/Unix** environments. Automated test execution, security scans, and deployment workflows via **GitHub Actions** pipelines.
- **Contract Testing & Modern Frontend Collaboration:** Defined strict OpenAPI/REST API schemas, collaborating with frontend engineers building **Vue.js** and React portals for Persons with Diabetes (PwDs). Maintained comprehensive unit and integration suites with **Pytest** and Jest to ensure 100% compliance with critical medical device data contracts.
- **Production Observability & Diagnostics:** Configured comprehensive monitoring, synthetic alerts, and structured audit logs across microservices using **AWS CloudWatch** and Linux CLI diagnostic scripts, ensuring zero-downtime availability for mission-critical health telemetry.

Project: CareExpand (EMR Platform)

Role: Senior Backend Engineer (Python & Microservices)

Stack: Python, NestJS, TypeScript, MongoDB (MongoEngine), MySQL, Sequelize, AWS (Lambda, API Gateway, SQS/SNS, EventBridge, CloudWatch), Docker & Docker Compose, GitHub Actions, Auth0 (OAuth 2.0, JWT, RBAC), Pytest, Vue.js, Linux/Unix

- **Microservices Modernization & Service Design:** Co-led the re-architecture of legacy monolithic clinical systems into decoupled **Python** and NestJS microservices, establishing structured technical designs and versioned REST APIs that improved platform uptime to 99.9% and eliminated architectural debt.
- **Hybrid Data Layer & Document Optimization:** Architected HIPAA-aligned data persistence by pairing relational patient registries in MySQL with flexible document schemas in **MongoDB** via **MongoEngine** to handle dynamic clinical notes, encounters, and audit trails; applied index tuning and query profiling to reduce complex report generation latency by 35%.
- **Identity, Access Control & API Security:** Engineered strict enterprise access controls using **Auth0**, implementing **OAuth 2.0/OIDC** protocols, **JWT** validation, token expiration handling, and multi-tier **RBAC** to restrict sensitive Electronic Medical Record (EMR) data access across doctors, nurses, and billing staff.
- **Asynchronous Integration & Serverless Tasks:** Built asynchronous background job pipelines utilizing **AWS EventBridge**, **SQS**, and **SNS** to process high-volume lab results and outbound notifications; deployed **AWS Lambda** functions behind **API Gateway** for scheduled data normalization jobs.
- **Docker Workflows & CI/CD Pipelines:** Containerized backend services with multi-stage **Docker** configurations and **Docker Compose** to maintain strict environment parity between local development and CI runners. Authored automated **GitHub Actions** workflows for continuous linting, security scanning, and test gates.
- **Contract Testing & Frontend Collaboration:** Formulated strict OpenAPI/REST contracts and consumer-driven tests, partnering with frontend teams developing **Vue.js** clinical dashboards to ensure robust type safety and zero integration drift. Standardized test coverage (>85%) with **Pytest** and Jest integration suites.

Mobile Programming Pvt. Ltd. | Software Developer | Sept 2021 – Sept 2023

Project: Core E-commerce Microservices | **Client:** CaratLane

Role: Senior Backend Engineer (Python, Node.js & Cloud)

Stack: Python, Node.js, Express, GraphQL, MongoDB (MongoEngine), MySQL, Redis, AWS (API Gateway, Lambda, EventBridge, SQS/SNS, Redshift, CloudWatch), Elasticsearch, Docker & Docker Compose, GitHub Actions, Auth0 (OAuth 2.0, JWT, RBAC), Pytest, Vue.js, React, Linux/Unix

- **Hybrid Microservices & Event-Driven Backbone:** Engineered and versioned high-throughput REST and GraphQL APIs across **Python** and Node.js microservices, serving mobile, web, and internal client applications. Built asynchronous event pipelines utilizing **AWS EventBridge**, **SQS**, and **SNS** to handle order lifecycles, payment webhooks, and flash-sale inventory updates seamlessly.
- **Document & Relational Persistence Layer:** Integrated **MongoDB** using **MongoEngine** alongside MySQL to handle flexible catalog schemas and high-velocity interaction logs, executing index tuning, sharding considerations, and query optimization to keep database read latencies low during high-traffic promotional surges.
- **Serverless Compute & Enterprise Security:** Deployed **AWS Lambda** serverless microservices in **Python** behind **API Gateway** for bursty e-commerce workflows and automated image/asset processing. Centralized IAM using **Auth0**, securing external and inter-service communications via **OAuth 2.0/OIDC**, JWT validation, token lifecycle handling, and granular **RBAC** policies.
- **Containerization & CI/CD Pipelines:** Standardized local development environments using multi-stage **Docker** builds and **Docker Compose**, ensuring parity between developer workstations and staging servers under **Linux/Unix**. Engineered automated **GitHub Actions** CI/CD pipelines to run vulnerability audits, linter checks, and deployment triggers.
- **Search Optimization & Data Orchestration:** Tuned **Elasticsearch** clusters and implemented a unified **GraphQL** aggregation layer, eliminating frontend over-fetching by 40% and accelerating product discovery across a high-traffic jewelry inventory.
- **API Contract Testing & Frontend Collaboration:** Established consumer-driven API contract testing and continuous test automation using **Pytest** and Jest to maintain >85% code coverage. Collaborated closely with frontend engineers building **Vue.js** and **React** store experiences to align schema definitions and eliminate API payload drift.
- **Analytics Pipeline & Enterprise Observability:** Architected CDC/ETL synchronization pipelines moving transactional records from relational stores into **AWS Redshift** for near real-time BI reporting; automated monitoring, alerting, and log tracking via **AWS CloudWatch** and custom shell diagnostics.

Project: SAP Value Lifecycle Manager (VLM) | Client: SAP

Stack: Node.js, Express, SAP HANA DB

- **Enterprise Logic:** Engineered HANA DB stored procedures to automate complex value-realization calculations for global enterprise clients.

Project: Smart Restaurant System (POS)

Stack: Node.js, Express, Parse Server, MongoDB

- **BaaS Implementation:** Leveraged Parse Server to rapidly deploy a secure, real-time POS system for high-volume restaurant environments.

Consortium of Institutions of Higher Learning | Software Developer | July 2017 – Aug 2021

- Developed web applications using Node.js and JavaScript to streamline institutional workflows.
- Implemented secure authentication and optimized database queries for student management systems.

GoLive Gaming Solutions Pvt. Ltd. | Game Developer | Sep 2016 – Apr 2017

- Developed interactive game logic and backend services using JavaScript and associated frameworks.
-

EDUCATION

- **Master of Technology (IT)** | JNTUH, Hyderabad | 2015 – 2017
- **Bachelor of Technology (Civil Engineering)** | St. Mary's Group of Institutions | 2010 – 2014